EST II — MATH LEVEL 2 • UNIT 6.01

C Permutations, Combinations & Conditional Probability

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 A
Q6 A	Q7 A	Q8 B	Q9 C	Q10 B
Q11 D	Q12 C	Q13 A	Q14 A	Q15 B
Q16 C	Q17 B	Q18 D	Q19 B	Q20 A
Q21 B	Q22 A	Q23 A	Q24 A	Q25 D

Permutations, Combinations & Conditional Probability Practice Exercises

1 MCQ

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7+4i}{2-i}$?

A $2 + 3i$ ✓

B $2 - 3i$

C $\frac{10+15i}{3}$

D $\frac{18+15i}{5}$

WORKED SOLUTION & VERIFICATION

To simplify the quotient of two complex numbers, multiply both the numerator and the denominator by the complex conjugate of the denominator, which is $2 + i$:

$$\frac{7+4i}{2-i} = \frac{(7+4i)(2+i)}{(2-i)(2+i)}$$

Expand the numerator using the distributive property (FOIL):

$$(7+4i)(2+i) = 14 + 7i + 8i + 4i^2$$

Since $i^2 = -1$:

$$14 + 15i + 4(-1) = 14 - 4 + 15i = 10 + 15i$$

Expand the denominator using the difference of squares:

$$(2-i)(2+i) = 2^2 - i^2 = 4 - (-1) = 5$$

Now, divide each term in the numerator by the denominator:

$$\frac{10+15i}{5} = \frac{10}{5} + \frac{15i}{5} = 2 + 3i$$

Common Misconceptions:

- Choice B occurs from an arithmetic sign error in the imaginary term.
- Choice C occurs if the student mistakenly calculates the denominator as $2^2 - 1^2 = 3$ by forgetting that $-i^2 = -(-1) = +1$.
- Choice D occurs if the student takes $i^2 = 1$, yielding $14 + 4 = 18$ in the numerator.

2 MCQ

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?

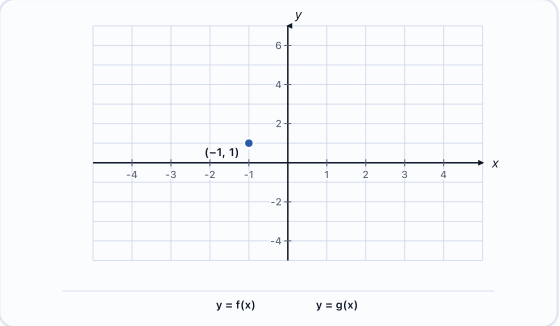
- **A** 0 ✓
- **B** 2
- **C** -1
- **D** 3

WORKED SOLUTION & VERIFICATION

The value of $f(1)$ corresponds to the y-coordinate of the point on the graph where the x-coordinate is 1. Looking at the provided graph, the point $(1, 0)$ is marked. Therefore, $f(1) = 0$.

3 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?



- **A** (1, 0) and (3, 0) ✓
- **B** (0, 1) and (0, 3)
- **C** (-1, 0) and (-3, 0)
- **D** (1, 0) and (-3, 0)

WORKED SOLUTION & VERIFICATION

The x-intercepts of a function occur where the function's value is zero, i.e., $f(x) = 0$. \nWe need to solve the quadratic equation $x^2 - 4x + 3 = 0$. \nThis equation can be factored as $(x - 1)(x - 3) = 0$. \nTherefore, the solutions are $x = 1$ and $x = 3$. \nThis means the x-intercepts are $(1, 0)$ and $(3, 0)$.

4 MCQ

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

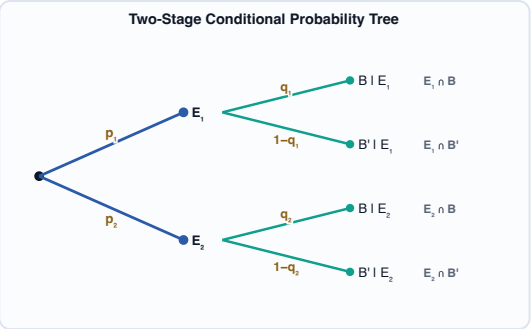
- **A** [3, 4] ✓
- **B** [5, 8]
- **C** [-3, -4]
- **D** [4, 3]

WORKED SOLUTION & VERIFICATION

The components of a vector $\mathbf{v}$ represented by a directed line segment from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$ are given by $\mathbf{v} = [x_2 - x_1, y_2 - y_1]$. In this case, $P = (1, 2)$ and $Q = (4, 6)$. Therefore, the components of $\mathbf{v}$ are $[4 - 1, 6 - 2] = [3, 4]$.

5 MCQ

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?



- **A** $\frac{3}{5}$ ✓
- **B** $\frac{2}{5}$
- **C** $\frac{1}{2}$
- **D** $\frac{3}{4}$

WORKED SOLUTION & VERIFICATION

The total number of marbles in the bag is the sum of red and blue marbles: $12 + 8 = 20$ marbles. The number of favorable outcomes (drawing a red marble) is 12. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of drawing a red marble is $P(\text{red}) = \frac{\text{Number of red marbles}}{\text{Total number of marbles}} = \frac{12}{20}$. This fraction can be simplified by dividing both the numerator and the denominator by their greatest common divisor, which is 4: $\frac{12 \div 4}{20 \div 4} = \frac{3}{5}$. Thus, the probability of drawing a red marble is $\frac{3}{5}$.

6 MCQ

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

- **A** 14 ✓
- **B** 17
- **C** 8
- **D** 11

WORKED SOLUTION & VERIFICATION

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n-1)d$. In this problem, we are given $a_1 = 5$ and $d = 3$. We need to find the 4th term, so $n = 4$. Plugging these values into the formula, we get: $a_4 = 5 + (4-1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$. Therefore, the 4th term of the sequence is 14.

7 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

- **A** 96 ✓
- **B** 192
- **C** 98
- **D** -96

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 6(2)^4 = 96$. Therefore, the verified correct option is 96.

8 MCQ

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.

A 0

B -2 ✓

C 2

D 1

WORKED SOLUTION & VERIFICATION

The general form of the quadratic function is $y = ax^2 + bx + c$. We are given three points that lie on the graph: $(-1, 5)$, $(1, 1)$, and $(3, 5)$.

Substituting these points into the equation, we get a system of three linear equations with three unknowns (a, b, and c):

1. For $(-1, 5)$: $a(-1)^2 + b(-1) + c = 5 \Rightarrow a - b + c = 5$
2. For $(1, 1)$: $a(1)^2 + b(1) + c = 1 \Rightarrow a + b + c = 1$
3. For $(3, 5)$: $a(3)^2 + b(3) + c = 5 \Rightarrow 9a + 3b + c = 5$

We can solve this system of equations. Let's subtract equation (1) from equation (2):

$(a + b + c) - (a - b + c) = 1 - 5$

$2b = -4$

$b = -2$

Alternatively, notice that the y-values for $x = -1$ and $x = 3$ are the same (both are 5). This indicates that the axis of symmetry of the parabola is exactly in the middle of $x = -1$ and $x = 3$. The x-coordinate of the vertex (axis of symmetry) is given by $-\frac{b}{2a}$.

Axis of symmetry: $x = \frac{-1+3}{2} = \frac{2}{2} = 1$

So, $-\frac{b}{2a} = 1$, which implies $b = -2a$.

Now, substitute this into equation (2): $a + b + c = 1$

$a + (-2a) + c = 1$

$-a + c = 1 \Rightarrow c = a + 1$

Substitute $b = -2a$ and $c = a + 1$ into equation (1):

$a - (-2a) + (a + 1) = 5$

$a + 2a + a + 1 = 5$

$4a + 1 = 5$

$4a = 4$

$a = 1$

Now we can find b using $b = -2a$:

$b = -2(1)$

$b = -2$

To confirm, we can also find c: $c = a + 1 = 1 + 1 = 2$. The function is $y = x^2 - 2x + 2$.

Let's check the points:

For $(-1, 5)$: $(-1)^2 - 2(-1) + 2 = 1 + 2 + 2 = 5$ (Correct)

For $(1, 1)$: $(1)^2 - 2(1) + 2 = 1 - 2 + 2 = 1$ (Correct)

For $(3, 5)$: $(3)^2 - 2(3) + 2 = 9 - 6 + 2 = 5$ (Correct)

The value of b is -2.

9 MCQ

A survey was conducted on a group of 100 students regarding their preferences for two subjects, Mathematics (M) and Science (S). The results are summarized in the two-way table below, where some values are unknown. If a student is chosen at random from this group, the probability that the student prefers Mathematics but not Science, $P(M \cap \neg S)$, is equal to the probability that the student prefers Science but not Mathematics, $P(\neg M \cap S)$. What is the value of x?

	Science (S)	Not Science (¬S)	Total
Mathematics (M)	25	x	45
Not Math (¬M)	y	35	55
Total	45	z	100

A 30

B 15

C 20 ✓

D 25

WORKED SOLUTION & VERIFICATION

From the first row (Mathematics), the total number of students who prefer Mathematics is 45, so:

$$25 + x = 45 \implies x = 20.$$

Since $P(M \cap \neg S) = P(\neg M \cap S)$, we also have $y = x = 20$, which matches the first column total $25 + 20 = 45$ and the second row total $20 + 35 = 55$.

10 MCQ

A manufacturer produces light bulbs. The probability that a randomly selected light bulb is defective is p. The production process is designed such that the probability of producing exactly k defective bulbs in a batch of $n = 5$ bulbs follows the binomial probability formula: $P(X = k) = \binom{n}{k}p^k(1-p)^{n-k}$. It is known that the probability of finding exactly 2 defective bulbs in a batch of 5 is $\frac{10}{32}$. Find the value of p.

A 0.8

B 0.5 ✓

C 0.75

D 0.2

WORKED SOLUTION & VERIFICATION

Let X be the number of defective bulbs in a batch of $n = 5$. We are given that X follows a binomial distribution with parameters $n = 5$ and probability of success (a bulb being defective) p. The probability of exactly k defective bulbs is $P(X = k) = \binom{n}{k}p^k(1-p)^{n-k}$.

We are given that the probability of finding exactly 2 defective bulbs in a batch of 5 is $\frac{10}{32}$. Therefore, $P(X = 2) = \binom{5}{2}p^2(1-p)^{5-2} = \frac{10}{32}$.

First, calculate the binomial coefficient: $\binom{5}{2} = \frac{5!}{2!(5-2)!} = \frac{5!}{2!3!} = \frac{5\times4}{2\times1} = 10$.

So, the equation becomes: $10p^2(1-p)^3 = \frac{10}{32}$.

Divide both sides by 10: $p^2(1-p)^3 = \frac{1}{32}$.

We need to find a value of p that satisfies this equation. Let's test the given options:

If $p = 0.2$:

$(0.2)^2(1-0.2)^3 = (0.04)(0.8)^3 = 0.04 \times 0.512 = 0.02048$. $\frac{1}{32} = 0.03125$. This is not correct.

If $p = 0.5$:

$(0.5)^2(1-0.5)^3 = (0.25)(0.5)^3 = 0.25 \times 0.125 = 0.03125$. $\frac{1}{32} = 0.03125$. This is correct.

If $p = 0.8$:

$(0.8)^2(1-0.8)^3 = (0.64)(0.2)^3 = 0.64 \times 0.008 = 0.00512$. This is not correct.

If $p = 0.75$:

$(0.75)^2(1-0.75)^3 = (0.5625)(0.25)^3 = 0.5625 \times 0.015625 = 0.0087890625$. This is not correct.

Thus, the value of p is 0.5.

11 MCQ

A bag contains 5 red marbles and 3 blue marbles. If a marble is drawn at random, what is the probability that it is blue?

A $\frac{5}{8}$

B $\frac{3}{5}$

C $\frac{5}{3}$

D $\frac{3}{8}$ ✓

WORKED SOLUTION & VERIFICATION

The total number of marbles in the bag is the sum of red and blue marbles: $5 + 3 = 8$. The number of favorable outcomes (drawing a blue marble) is 3. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of drawing a blue marble is $\frac{\text{Number of blue marbles}}{\text{Total number of marbles}} = \frac{3}{8}$.

12 MCQ

A factory produces two types of components, A and B. Components of type A have a defect rate of 3%, and components of type B have a defect rate of 5%. A batch of 100 components is produced, consisting of N components of type A and $100 - N$ components of type B. Let X be the random variable representing the total number of defective components in the batch. If the expected number of defective components is $E[X] = 4.2$, find the probability that a randomly selected component from this batch is defective, given that it is of type A.

A 0.036

B 0.042

C **0.030** ✓

D 0.050

WORKED SOLUTION & VERIFICATION

Let X_A be the number of defective components of type A, and X_B be the number of defective components of type B. The total number of defective components is $X = X_A + X_B$.
The expected number of defective components of type A is $E[X_A] = N \times 0.03$.
The expected number of defective components of type B is $E[X_B] = (100 - N) \times 0.05$.
The total expected number of defective components is $E[X] = E[X_A] + E[X_B]$.
We are given $E[X] = 4.2$.
So, $N \times 0.03 + (100 - N) \times 0.05 = 4.2$.
$0.03N + 5 - 0.05N = 4.2$.
$5 - 0.02N = 4.2$.
$0.02N = 5 - 4.2$.
$0.02N = 0.8$.
$N = \frac{0.8}{0.02} = 40$.
Therefore, there are 40 components of type A and $100 - 40 = 60$ components of type B.
The question asks for the probability that a randomly selected component from this batch is defective, given that it is of type A. This is simply the defect rate of type A components.
Probability (Defective | Type A) = Defect rate of Type A = 0.03.

13 MCQ

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

A **1/3** ✓

B 1/6

C 1/2

D 2/3

WORKED SOLUTION & VERIFICATION

The possible outcomes when rolling a fair six-sided die are {1, 2, 3, 4, 5, 6}. The total number of possible outcomes is 6.
We are interested in the event of rolling a number greater than 4. The numbers in the set of outcomes that are greater than 4 are {5, 6}.
There are 2 favorable outcomes.
The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes.
Probability (rolling > 4) = (Number of favorable outcomes) / (Total number of possible outcomes) = 2 / 6 = 1/3.

14 MCQ

A fair six-sided die is rolled. Find $P(\text{rolling a } 3)$.

- **A** $\frac{1}{6}$ ✓
- **B** $\frac{5}{6}$
- **C** $\frac{1}{3}$
- **D** $\frac{1}{2}$

WORKED SOLUTION & VERIFICATION

One favorable outcome out of 6 equally likely outcomes: $P = \frac{1}{6}$.

15 MCQ

A fair six-sided die is rolled. Find $P(\text{rolling a } 4)$.

- **A** $\frac{1}{3}$
- **B** $\frac{1}{6}$ ✓
- **C** $\frac{1}{2}$
- **D** $\frac{5}{6}$

WORKED SOLUTION & VERIFICATION

One favorable outcome out of 6 equally likely outcomes: $P = \frac{1}{6}$.

16 MCQ

In an applied problem, a fair six-sided die is rolled. Find $P(\text{rolling a } 2)$.

- **A** $\frac{5}{6}$
- **B** $\frac{1}{3}$
- **C** $\frac{1}{6}$ ✓
- **D** $\frac{1}{2}$

WORKED SOLUTION & VERIFICATION

One favorable outcome out of 6 equally likely outcomes: $P = \frac{1}{6}$.

17 MCQ

Suppose a fair six-sided die is rolled. Find $P(\text{rolling a } 1)$.

- **A** $\frac{1}{3}$
- **B** $\frac{1}{6}$ ✓
- **C** $\frac{1}{2}$
- **D** $\frac{5}{6}$

WORKED SOLUTION & VERIFICATION

One favorable outcome out of 6 equally likely outcomes: $P = \frac{1}{6}$.

18 MCQ

An observed fair six-sided die is rolled. Find $P(\text{rolling a } 5)$.

A $\frac{5}{6}$

B $\frac{1}{3}$

C $\frac{1}{2}$

D $\frac{1}{6}$ ✓

WORKED SOLUTION & VERIFICATION

One favorable outcome out of 6 equally likely outcomes: $P = \frac{1}{6}$.

19 MCQ

Consider a fair six-sided die is rolled. Find $P(\text{rolling a } 6)$.

A $\frac{1}{3}$

B $\frac{1}{6}$ ✓

C $\frac{1}{2}$

D $\frac{5}{6}$

WORKED SOLUTION & VERIFICATION

One favorable outcome out of 6 equally likely outcomes: $P = \frac{1}{6}$.

20 MCQ

A fair six-sided die is rolled. Find $P(\text{rolling a number less than } 3)$.

A $\frac{1}{3}$ ✓

B $\frac{1}{2}$

C $\frac{2}{3}$

D $\frac{1}{6}$

WORKED SOLUTION & VERIFICATION

2 of the 6 outcomes satisfy this: $P = \frac{1}{3}$.

21 MCQ

A fair six-sided die is rolled. Find $P(\text{rolling a number greater than } 4)$.

A $\frac{1}{2}$

B $\frac{1}{3}$ ✓

C $\frac{2}{3}$

D $\frac{1}{6}$

WORKED SOLUTION & VERIFICATION

2 of the 6 outcomes satisfy this: $P = \frac{1}{3}$.

22 MCQ

If $P(E) = \frac{14}{19}$, find $P(E')$.

- **A** $\frac{5}{19}$ ✓
- **B** $\frac{14}{19}$
- **C** $\frac{6}{19}$
- **D** 1

WORKED SOLUTION & VERIFICATION

$P(E') = 1 - P(E) = 1 - \frac{14}{19} = \frac{5}{19}$.

23 MCQ

Suppose that $P(E) = \frac{3}{11}$, find $P(E')$.

- **A** $\frac{8}{11}$ ✓
- **B** $\frac{3}{11}$
- **C** $\frac{9}{11}$
- **D** 1

WORKED SOLUTION & VERIFICATION

$P(E') = 1 - P(E) = 1 - \frac{3}{11} = \frac{8}{11}$.

24 MCQ

Assuming that $P(E) = \frac{4}{19}$, find $P(E')$.

- **A** $\frac{15}{19}$ ✓
- **B** $\frac{16}{19}$
- **C** 1
- **D** $\frac{4}{19}$

WORKED SOLUTION & VERIFICATION

$P(E') = 1 - P(E) = 1 - \frac{4}{19} = \frac{15}{19}$.

25 MCQ

Under the condition that $P(E) = \frac{8}{9}$, find $P(E')$.

- **A** $\frac{8}{9}$
- **B** $\frac{2}{9}$
- **C** 1
- **D** $\frac{1}{9}$ ✓

WORKED SOLUTION & VERIFICATION

$P(E') = 1 - P(E) = 1 - \frac{8}{9} = \frac{1}{9}$.